A Multi-Dimensional Pipeline for Holistic Prosodic Visualization: Integrating f_0, Intensity, and Syllabic Rate using *VIP2VIP*

Federico Lo Iacono[1,2], *Antonio Romano*[1]

[1] University of Turin, Laboratorio di Fonetica Sperimentale – "Arturo Genre"
[2] University of Genoa
{firstname.lastname}@unito.it

Abstract

Most visualizations used in speech prosody research privilege a single acoustic dimension (typically f_0), leaving key correlates of prominence, phrasing, and rhythm only indirectly inferable. We present *VIP2VIP – Visualize Intonation and Prosody for Voices of Italian Poets*: a new, open pipeline for *multi-dimensional* intonation visualization that renders (I) an f_0 contour smoothed by LOWESS or GAM, (II) instantaneous *intensity* (as line thickness), and (III) *syllabic rate* (SR; syllables/s) projected onto the f_0 trace as a perceptually ordered colour ramp. The visualization operates over light, reproducible annotations and is robust to micro-prosodic noise. It is intended as a descriptive tool that remains compatible with superpositional theories: global phrase-level trends and local accentual movements can be inspected together with temporal density and energy. We detail the method and its User Interface, illustrate analyses on Italian poetic read speech, and compare this approach to f_0-only and stylization-based tools. We suggest that compressing these three dimensions into a single coordinated display may facilitate hypothesis generation, while remaining usable alongside formal modelling frameworks such as Fujisaki's or Grønnum's approach.

Index Terms: Intonation, Superpositional approach, Prosody, Stylistic features, Holistic analysis of Prosody

1. Introduction

Visual analysis of intonation traditionally foregrounds the f_0 track (raw or stylized), with intensity and segmental tiers in separate panes. Landmarks include ToBI-style tiers, stylization models (MOMEL/INTSINT, *Prosogram*), and parametric or polynomial summaries. These approaches advance comparability and cognitive plausibility, yet the visualization remains univariate at each instant, leaving intensity and temporal density inferred rather than co-varying with f_0 [1-3]. An f_0-only plot obscures two crucial dimensions: energy and tempo. Intensity shapes emphasis and phrasing but is rarely co-visualized with f_0; syllabic rate (SR) cues function and interacts with prominence and pragmatic structure. Without integration, analysts rely on shortcuts or post hoc statistics, hiding local trade-offs (e.g., broader excursions with slower tempo) and misattributing prominence to pitch alone. There are important precedents that gesture toward the integration of these dimensions. *PraaPer* pairs f_0 with periodic-energy proxies and scripted graphics [4]. AMPER conceptualizes intonation as a global, continuous phenomenon and cautions against arbitrary segmentation, advocating preservation of the full contour while superimposing temporal scaffolding (e.g., syllables, labels) and co-varying intensity/SR fields to permit multiscale interpretation without forced discretization [5-7]. Our pipeline is informed by superpositional accounts in which f_0 is often analysed as a slowly varying phrase trend plus local accent movements and segmental perturbations. Mapping intensity and SR onto the f_0 trace is intended to make visible cases where accent-like movements appear to be energy- and timing-supported and cases where phrase-level drift proceeds with little such support. This logic spans (I) command–response formulations (Fujisaki's phrase/accent commands driving second-order systems over a base log-f_0), (II) parametric separations of phrase-level and localized components, and (III) unit-superpositional systems that learn context-dependent prominence weights and rhythmic scopes [8-18]. Estimation work shows that relaxing accent-configuration constraints and using informed initialization improve robustness; our script makes these choices explicit by exposing smoothing and unit selection. Following Grønnum's layered, superpositional view, global textual organization constrains local realizations: utterances descend within a textual contour, phrases nest within utterances, and lower-level events are subordinate to higher-level structure. This textuality, for instance, is crucial to poetic speech (the register we focus on here, mainly drawing on Italian read-aloud speech), where lines, stanzas, and enjambments scaffold prosodic expectations in reading aloud [19-22]. Grønnum also emphasizes temporal planning (slope compression and expansion, look-ahead, and rate modulation tied to upcoming goals), implying that prominence and boundary perception result from co-evolving pitch, energy, and timing, not pitch alone. Our tri-modal overlay operationalizes this stance by showing verse- and stanza-level organization as coordinated patterns of f_0 curvature, thickness, and SR gradients. Against this backdrop, the paper addresses the gap between cognitively plausible models and predominantly univariate graphics. Our research questions are whether a tri-modal overlay clarifies local trade-offs among pitch, energy, and tempo; better supports prominence analysis beyond f_0; and how these joint cues pattern across prosodic units. To address these questions, we introduce a reproducible pipeline that (I) overlays intensity and SR on a smoothed f_0 contour, (II) offers LOWESS or GAM smoothing with syllable-tuned flexibility, (III) reads TextGrids, and (IV) outputs per-unit summaries.

2. Theoretical background

We propose considering intonation holistically and superpositionally rather than discretely. In Grønnum's account, a gradually descending textual contour spans multi-utterance stretches; utterance contours descend within it, and prosodic phrases align to these trends. Local stress-group patterns are scaled and timed by higher-level structure, and speakers use look-ahead, adjusting slopes and timing to remaining structure. This hierarchy motivates an analysis that privileges continuous

contours and relational scaling over categorical partitioning. A multi-dimensional perspective follows naturally. While f_0 is a convenient carrier, prominence and phrasing arise from convergent cues (e.g., pre-boundary lengthening with energy drop, crescendos with rising movement). Integrating SR and intensity with f_0 reveals superposition directly – global descent with thinning versus accent curvature with thickening and local SR cooling – supporting interpretation that would be obscured in f_0-only views. Consequently, our pipeline adopts a superpositional framework. It was developed and validated on Italian read-aloud recordings, but its central commitments—preserving continuous contours and using within-unit relational scaling—do not presuppose language-specific prosodic categories and are intended to generalize cross-linguistically. In Fujisaki's command–response model, log-f_0 sums phrase- and accent-command outputs over a speaker-specific base; analysis-by-synthesis shows phrase-command timing tracks hierarchical phrasing, whereas accent timing/amplitude track local prominence, focus, and lexical/tonal structure in read and spontaneous speech [8-12]. Recent fits avoid rigid accent templates, seeding with linguistically motivated cues (stressed/focused syllables, phrase edges) [13]. Visualizations that highlight f_0 movements with energy investment and local slowing provide theory-neutral, perceptually grounded cues for estimation. The tri-modal overlay offers visual cues to where putative categories appear energy- and time-licensed through line thickness and local SR cooling. A parallel line, WSFC, models prosody as a weighted sum of functional contours anchored to rhythmic units [14-17]; because these contours are multi-parametric, co-registering energy and timing with f_0 suits hypothesis generation and qualitative checks before fitting. Within this general superpositional orientation, AMPER advocates for a global treatment and joint analysis of accent with intonation; we adopt this stance by exposing multiple control dimensions, allowing distinctions to emerge from the patterned co-variation of f_0, intensity, and SR [5-7]. Preserving the full contour while attending to co-varying temporal and energetic cues aligns with AMPER's continuous view. Furthermore, *PraaPer* emphasizes periodic energy salience and reproducible, scripted plots; using periodic energy as the thickness driver ties visual prominence to voicing quality and improves cross-session comparability [4]. These considerations inform our choice to co-register energy and timing with f_0 to make multimodal cueing explicit. For instance, *WinPitch* operationalizes a perceptual perspective with a continuous, parametric representation: pitch tracking, glissando thresholds, and domain-sensitive diagnostics support incremental judgments and test whether threshold-crossing movements are energetically supported and temporally licensed. As an alternative to categorical discretization, *WinPitch* offers a configurational approach [23-25]. Our overlay builds on this by making licensing conditions visible and auditable. Other frameworks complement this landscape. MOMEL/INTSINT and *Prosogram* compress f_0 into plausible targets that facilitate comparison and transcription. These models clarify cross-speaker alignment and support symbolic tiering, but the stylized line is typically shown alone, with intensity and tempo elsewhere. However, discrete, symbol-centric approaches are less suitable for our aims. Perceptually, category-level rises/falls are those exceeding glissando thresholds within stress-group and phrase domains, yet categorical inventories underplay continuous scaling, temporal licensing, and energy budgeting. By binding energy and tempo to the pitch trace, our model preserves continuity, supports superpositional interpretation, and offers a Gestalt-based view of melodic rhythm [26-27]. For poetry read aloud, this yields practical diagnostics: textual descent may level off mid-text, internal units can be less slanted than terminal declaratives, and phrasing may track syntactic, semantic, or metrical coherence. Beyond accent and phrase commands, the global melodic–rhythmic surface reflects metrically salient positions, caesuras, and verse semantics. This links perception-oriented diagnostics and superpositional modelling to a visualization that aligns qualitative interpretation with model-based inference.

2.1. Compatibility with superpositional models

Superpositional accounts model log-f_0 as the sum of phrase-level and accent-level contributions plus segmental micro-perturbations. Such modelling lies beyond the scope of the present paper; instead, the visualization is designed to be interpretable in ways that are consistent with these frameworks. In this perspective, localized curvature combined with line thickening and SR slowdown is one possible diagnostic of accent-like control, whereas gentler curvature with thinning is one possible cue to phrase-level drift. These visual diagnostics could inform subsequent choices about command timing and amplitude in command–response models, or about contour density in WSFC-style approaches. In estimation work, linguistically seeded, non-templatic fits have been shown to be robust; our overlay is meant to support such practices by providing a transparent, perception-oriented view of where strong movements coincide with energy investment and local slowing. We treat the overlay as model-compatible rather than model-bound: it offers auditable evidence that may help analysts formulate expectations about phrase- and accent-level controls, which can then be tested in formal superpositional fits [8–18]. Under Grønnum's hierarchy, the overlay separates levels visually (global text/utterance slopes vs. local stress-group patterns), which we take as a convenient descriptive complement to, rather than a replacement for, explicit modelling of hierarchical superposition.

3. Methodology

The open pipeline runs in *Python* with *Matplotlib* graphics, using *Jupyter Notebook*; f_0 and intensity are extracted via *Parselmouth* (*Praat* backend). Smoothing is provided by *statsmodels* LOWESS or GAM (*GLMGam* with B-splines). Annotation parsing is handled by *PraatIO*. The visualization function accepts tier names (CP, *syll*, *phon*, INT, VS), excludes pause/noise intervals, colours the pitch trace by SR, scales line width by instantaneous intensity, and writes per-CP legends (mean f_0, mean syllable duration, SR, syllable count). All parameter choices (e.g., "LOWESS frac=x"; "GAM df, λ") are printed to the log for each CP [28-30]. The pipeline is designed to run on lightweight *Praat* TextGrids with the following tiers:

- **CP**: the working prosodic unit for visualization (e.g., "chunk" from pause to pause, or task-specific unit). Pauses are explicitly marked, but are excluded from processing.
- **syll**: syllable intervals (phonetic level), excluding pauses/noise; used for SR and labelling.
- **phon**: voiced phonetic intervals (SAMPA-like set including vowels and voiced sonorants); samples for f_0/intensity are taken only within these spans to limit microprosodic voiceless gaps.
- **INT**: intonational labels (e.g., continuations, interrogatives, focalization) which, if present, are printed on a lower track.

- **VS**: textual verse (when working with poetry).

Label conventions follow [31]. We reference a compact set of Italian language field labels used for intonational units [32], information structure, and pragmatics; our code reads them for plotting and does not constrain analytic choices. The pipeline is annotation-minimal: SR uses the *syll* tier only; f_0/intensity sampling requires a *phon* tier; CP windows drive plotting.

3.1. Acoustic extraction

Audio is read with *Parselmouth*. f_0 and intensity are queried at 5-ms steps only within voiced spans from the *phon* tier; this reduces artifacts at voiceless consonants and yields smoother instantaneous pairing of f_0 and intensity. A lightweight outlier filter (z-score threshold, default 2.5) trims extreme f_0 values without flattening natural excursions. Intensity values are carried forward unmodified (*Praat*'s to_intensity) [28].

3.2. Syllabic rate

We compute the syllabic rate (SR) at two scales. A global CP-level SR (syllables per second) is defined as the number of syllables fully contained in the CP divided by the CP duration; this value is reported in the plot legend. To capture local tempo dynamics, we also derive an instantaneous SR proxy from the inverse of each syllable's duration ($1/\Delta t$). These per-syllable speed values are attached to syllable centre times and linearly interpolated onto the f_0 sampling grid. After min–max normalization within CP, we map the resulting series to a perceptually ordered ramp so that green marks a slower local rate, yellow a mid-rate, and red a faster rate (relative to other segments within the same interpausal unit). Thus, the single f_0 polyline simultaneously encodes curvature (pitch), thickness (intensity), and hue (local and relative syllabic rate), allowing prominence, phrasing, and tempo to be read at a glance.

3.3. Smoothing

We implement two smoothing backends:

- LOWESS (default): robust local regression with bandwidth frac (0.05–0.80). LOWESS respects local curvature and resists outliers; it is well-suited for visualization that preserves accent-level wiggles [29].
- GAM: penalized cubic B-splines fit to normalized time with user-selectable degrees of freedom and penalty (λ). GAM affords global control over wiggliness and proves stable across variable CP durations [30]. A practical issue is choosing smoothness across CPs of different syllable counts. We implement Auto Flex. Given the number of syllables in the CP and a user "flex" target (0–120%), the system maps to a bounded df (cap ≈ 50) and exponentially decreasing λ ($\approx 0.8 \rightarrow 0.005$): long phrases are allowed more bends, while short phrases remain parsimonious. This maintains visual comparability and aligns with the superpositional intuition: intonative curve varies slowly; accents are preserved when warranted by syllabic density.

3.4. Visual encoding, user interface, and batch processing

Each interpausal unit (CP) is plotted in two vertically aligned panels sharing the x-axis.

- Upper panel (f_0): the smoothed f_0 trajectory is drawn as a polyline of short segments; line thickness encodes instantaneous intensity (thicker = louder), and colour encodes SR (from cool to warm as SR increases, using a perceptually ordered ramp). A dashed line marks mean f_0 within the CP; the legend reports mean f_0 (Hz), mean syllable duration (ms), SR (syll/s), and syllable count.
- Lower panel (INT labels): syllable boundaries and labels are printed at mid-interval positions; if present, INT labels are shown with vertical dashes.

This single-canvas display combines three prosodic dimensions and coordinates them without multiple subplots. Analysts read curvature, amplitude, and time density as an integrated gesture. In addition, the notebook User Interface allows *Batch* and *Interactive* modes with multi-file, per-CP selection. Presets provide f_0 ranges and default LOWESS bandwidths for male, female, and older voices; smoothing can be toggled between LOWESS and GAM, with Auto Flex controls visible for the latter. Status messages record the smoother and parameters used per CP, ensuring reproducibility. The pipeline allows saving all extracted indices in CSV format, enabling statistical analysis, and exporting the generated graphs as .png images.

4. Applications and scope

We present case studies showing how the tri-modal display reveals prominence, phrasing, and tempo patterns. The examples presented are derived from recordings housed in the *Voices of Italian Poets* oral archive [33]. The oral recitation of poetry is a distinctly stylized mode of oral expression that requires a multi-dimensional analytical approach for comprehensive characterization [19-22]. Claims that poetry reading is uniformly monotone (marked by unassertive, often suspended contours) are widespread. While such patterns are indeed observable (see Figs. 1-2 for relatively flat, suspended curves typical of *poetic declaratives* [19]), they do not exhaust the range of realizations. Many performances display more melodic, sinuous contours with clearly articulated phrasing that foregrounds metrical structure and places prominence at predictable loci (Fig. 3). From a superpositional perspective, examination of this intonation curve permits the analytical decomposition of the global f_0 movement into a underlying and general continuation function, which undergoes modulation through a scanning strategy that generates three rising pitch excursions aligned with the primary metrical stresses in Eugenio Montale's poetic recitation of one of his verses from *Meriggiare pallido e assorto* ('Noonday slumber pallid and rapt'): *ascol**ta**re tra i **pru**ni e gli s**ter**pi* ('listen among the brambles and brushwood'). Fig. 3 also illustrates the prosodic structure of a case of a two-element enumerative, which could demonstrate the superpositional nature of intonation: the initial rise on "*pru-*" anticipates the subsequent rise on "*ster-*" and is scaled to a lower f_0. Absent the second conjunct, "*pruni*" would likely have reached a comparably high, more suspensive peak; in the coordinated sequence, however, the rise on "*sterpi*" is compressed by the coordination pattern yet begins from an upstepped onset (i.e., a higher f_0 starting point) [7]. Moreover, the repertoire can span the language's ordinary intonational arc: for example, Fig. 4 shows a sequence of minor continuation (/ct/) followed by major continuation (/CT/), with the latter signalled by a rise in f_0. In this case, syllabic rate provides a decisive cue to function: a faster tempo at the onset of /ct/ gives way to slower, more suspended timing toward the end of /CT/, distinguishing the two continuation types and clarifying their pragmatic roles. A further intonational hallmark is the relative scarcity of fully assertive contours in medial positions, whereas stanza-final and poem-final units are pragmatically dominated by assertive patterns. Moreover, a multimodal visualization of

this kind enables an accurate depiction of stylistic phenomena such as final lengthening, which is widely used in poetry to highlight the salient *loci* of the text [34].

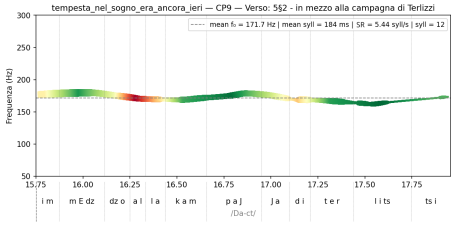

Fig. 1: CP realized like a /Da-ct/ ['In the middle of the Terlizzi countryside']

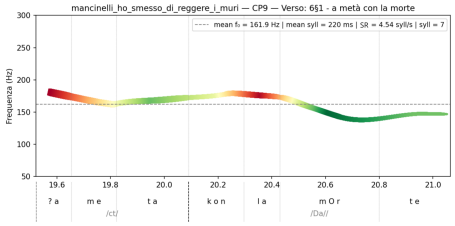

Fig. 2: CP realized like a /ct/ + /Da// ['Halfway to death']

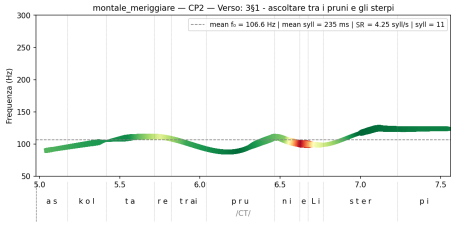

Fig. 3: CP realized like a /CT/ ['listen among the brambles and brushwood']

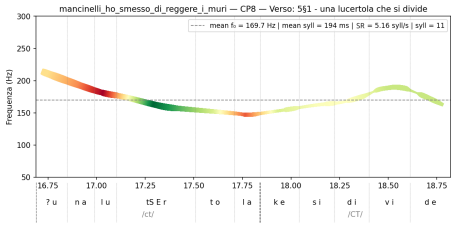

Fig. 4: CP realized like a /ct/ + /CT/ ['a split-tailed lizard']

5. Discussion

Traditional f_0-only plots force analysts to infer energy and tempo, while pane-separated displays add misalignment and cognitive load. Our visualization restores time-synchronous prosody: it shows emphatic (thick, cool) vs incidental (thin, warm) rises and whether boundaries combine pre-pause lengthening with energy decay. Explicit, documented smoothing (LOWESS/GAM) avoids ad hoc filtering and enables cross-file comparability via Auto Flex. The approach also could merge with *WinPitch*-style hypothesis testing and with *PraaPer*'s periodic-energy analysis, which can be plugged in as an alternative thickness metric to control for microphone or room effects while keeping annotations light. It is theory-compatible: in command–response models, the joint signature may help analysts hypothesize where phrase- and accent-level controls are likely to occur; in unit-superpositional accounts, SR colour and energy expenditure can be read as indicative of contour density and contextual prominence. In this sense, the overlay can serve as a descriptive front-end for parameter estimation and aligns with AMPER's commitment to analysing accent and intonation jointly without fragmenting the contour. Stylization systems layer on top: our display provides an information-rich front end with aligned targets for categorical decisions and corpus annotation (i.e., a layer for ToBI-style annotation). In practical terms, analysts gain a reproducible, interpretable view that preserves global shape while exposing interdependencies among pitch movement, energy, and timing that underwrite communicative function. Through Grønnum's lens, many pitch-only movements are text-conditioned realizations of a global descent, with SR modulation and energy budgeting equally explanatory to pitch. Verse-final cadences and stanza-internal levelling in poetry are therefore holistic outcomes of hierarchical planning, not merely local edge tones.

5.1. Limitations and future improvements

Our visualization inherits limits from the annotation and signal quality. Because SR is computed from syllable durations, merges/splits, and boundary-straddling in the *syll* tier directly bias the colour map; intensity is likewise affected by microphone distance, channel compression, and room acoustics. Within-file patterns remain informative, but cross-session use requires normalization. Smoothing also introduces variance: LOWESS bandwidth and GAM df/penalty shape curvature. Auto-Flex reduces arbitrariness, but settings should be reported and over-wiggling avoided. Genre matters: poetry's longer planning units can yield verse-level SR cooling and warming; conversational data call for denser syllabification and a more conservative reading of tempo. To improve robustness, we will adopt semitone scaling with speaker-range normalization for f_0, replace raw intensity with loudness-based energy, compute per-syllable energy area, and integrate voice quality indices (harmonicity/CPP, H1–H2, spectral slope) as thickness drivers or auxiliary channels, aligning with [4].

5.2. Script dissemination

The script is openly available on the first author's website: https://github.com/fedeloiac/vip-pipeline

6. Conclusions

We introduced a multi-dimensional visualization for speech prosody that *simultaneously* renders f_0, intensity, and syllabic rate on a single, time-aligned canvas. By encoding energy as thickness and tempo as colour *on the pitch line*, the method respects intonation's holistic nature and reduces reliance on discretionary segmentation. The pipeline is open and reproducible, and it is designed to remain compatible with superpositional theories and stylization or transcription frameworks. In applications to poetry and literary performance, it exposes style signatures, catches the deep melodic rhythm [26, 27], and clarifies prominence diagnostics that f_0-only plots obscure. We hypothesize that this visualization can reduce the cognitive load of multi-cue reasoning and support faster hypothesis formation. Future work will add normalization and loudness metrics, bridging exploratory graphics and formal parameter estimation. More broadly, the overlay is intended as a practical analytic tool inspired by the AMPER project and superpositional approaches in general, especially Grønnum's layered textuality and temporal planning. By binding energy and timing to f_0, it aims to make hierarchical superposition more easily inspectable and to provide a basis for future formal tests.

7. References

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